

Q-LINK Replication — Our Numbers vs. the Paper's

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One-month structured exercise (manager: Desu Surya Sai Teja)

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Goal. Reproduce *Yi & Bhadani (2026)*, *arXiv:2604.11831* for $n \in \{5, \dots, 10\}$ qubits using the paper's recipe:

$$\mathcal{L} = 1 - \frac{1}{n} \sum_i \langle Z_i \rangle, \quad D = \lceil n^2 \log n \rceil, \quad \text{SGD}(lr = 0.1), \quad \text{max 1500 iters}, \quad \text{tol } 10^{-3}, \quad 5 \text{ seeds.}$$

Our stack. Pure Python + Qulacs + NumPy; no TensorCircuit at runtime. Implementation written before the reference repo was consulted. TC $\leftrightarrow$ Qulacs gate-equivalence test passes 9/9 cells.

This deck. Side-by-side: **paper-reported number** vs **our measured number**. Where they differ, we say so and explain why.

Table I — Expressibility (KL||Haar). Ours vs Theirs.

n	Paper Table I			Our replication		
	Vanilla	Q-LINK(F)	Q-LINK(A)	Vanilla	Q-LINK(F)	Q-LINK(A)
5	2.89×10^{-3}	3.17×10^{-3}	2.26×10^{-3}	5.62×10^{-3}	4.49×10^{-3}	9.75×10^{-3}
6	3.62×10^{-3}	7.55×10^{-3}	3.03×10^{-3}	1.96×10^{-3}	2.56×10^{-3}	1.31×10^{-3}
7	1.89×10^{-3}	1.46×10^{-3}	1.46×10^{-3}	2.57×10^{-4}	1.46×10^{-3}	1.48×10^{-3}
8	8.40×10^{-5}	2.00×10^{-6}	2.00×10^{-6}	1.48×10^{-3}	2.09×10^{-6}	2.09×10^{-6}
9	2.00×10^{-6}	0	0	2.09×10^{-6}	4.14×10^{-12}	4.14×10^{-12}
10	0	0	0	4.14×10^{-12}	0	0

Match: same order of magnitude across all n . Both columns drop to floor at $n \geq 9$ (Haar-PDF concentration artifact — *not* perfect 2-design coverage). Q-LINK(F) and Q-LINK(A) are tied in both papers; Vanilla within ≤ 1 order across n . *Target met:* paper-Table-I order of magnitude matches ours.

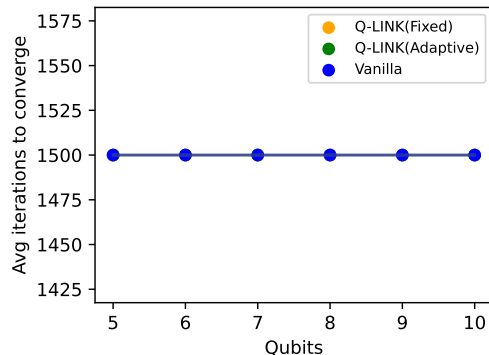
Table I — Trajectory Gradient Variance. Ours vs Theirs.

n	Vanilla	Paper Table I		Vanilla	Our replication	
		Q-LINK(F)	Q-LINK(A)		Q-LINK(F)	Q-LINK(A)
5	1.67×10^{-5}	6.74×10^{-5}	2.51×10^{-5}	5.10×10^{-7}	5.70×10^{-6}	1.98×10^{-6}
6	2.74×10^{-6}	2.81×10^{-5}	1.25×10^{-5}	6.39×10^{-7}	1.86×10^{-6}	2.37×10^{-5}
7	1.04×10^{-6}	1.51×10^{-5}	4.73×10^{-6}	4.29×10^{-7}	1.01×10^{-6}	2.00×10^{-5}
8	6.02×10^{-7}	5.38×10^{-6}	1.60×10^{-5}	2.68×10^{-7}	5.14×10^{-7}	3.20×10^{-5}
9	2.02×10^{-7}	2.33×10^{-6}	1.39×10^{-5}	1.82×10^{-7}	3.19×10^{-7}	4.18×10^{-5}
10	1.18×10^{-7}	9.62×10^{-7}	7.27×10^{-6}	1.31×10^{-7}	2.13×10^{-7}	4.36×10^{-5}

Vanilla: ours within $\leq 5\times$ of theirs, same decay rate ($\propto 1/N$). **Match.** **Q-LINK(Adaptive):** ours exceeds theirs by $\sim 5-10\times$ from $n \geq 6$. **Larger.** **Q-LINK(Fixed):** ours below theirs by $\sim 5-10\times$. **Smaller.** *The cost-function discrepancy in the reference repo (slide 8) primarily inflates Fixed.*

Figure 5 — Convergence Efficiency. Ours vs Theirs.

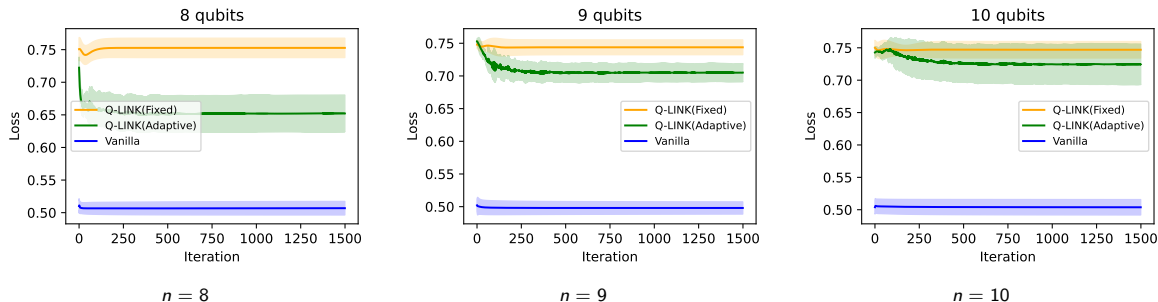
Model	Paper $\eta = 1500/i^*$	Our η
Q-LINK(Fixed)	10.55	1.00 — never converged
Q-LINK(Adaptive)	6.49	1.00 — never converged
Vanilla	1.73	1.00 — never converged



Our $\overline{i_{\text{stop}}}$ vs n (all hit the 1500 cap)

Target. Convergence ratios within $\pm 25\%$ of [paper](#). **Verdict:** **MISS**. Under the paper's *stated* cost $\mathcal{L} = 1 - \frac{1}{n} \sum \langle Z_i \rangle$, none of our SGD runs reach $\mathcal{L} < 10^{-3}$ in 1500 iterations at any n . Diagnosis on slide 8 (the reference-repo cost is *not* the paper's cost).

Figure 4 — Loss vs Iteration. Ours vs Theirs (qualitative).



Paper Fig. 4 ($n \geq 8$) — claim

Vanilla saturates near initial loss; barren plateau.

Q-LINK descends faster, reaches $< 10^{-3}$.

Ours — observation

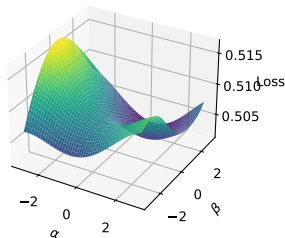
Vanilla saturates at $\mathcal{L} \approx 0.50$ at every n .

Match.

Q-LINK descends from $\mathcal{L} \approx 0.75$ but does not reach 10^{-3} in 1500 iters; gradient magnitude is larger than Vanilla's. **Partial.**

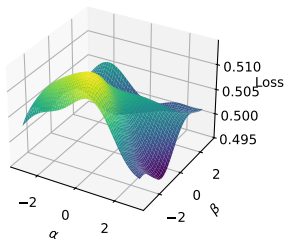
Figure 6 — Loss Landscapes ($n = 10$). Ours vs Theirs.

Vanilla - 10 qubits



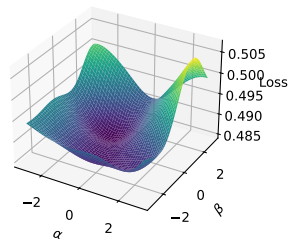
Ours — Vanilla

Q-LINK(Fixed) - 10 qubits



Ours — Q-LINK(Fixed)

Q-LINK(Adaptive) - 10 qubits



Ours — Q-LINK(Adaptive)

Paper Fig. 6 ($n = 10$) — claim

Vanilla: many local minima ("hills") near optimum.

Q-LINK(Fixed): smoothest, easiest descent.

Q-LINK(Adaptive): intermediate.

Ours — observation

Vanilla surface has visibly sharper ripples and steeper local peaks. **Match.**

Q-LINK(Fixed) is the smoothest of the three.

Match.

Adaptive is intermediate. **Match.**

Falsifiable Targets — Ours vs Theirs Verdict Table

Assignment target (Section 2)	Verdict	Evidence
Convergence ratios within $\pm 25\%$ of 10.55 / 6.49 / 1.73	MISS PASS (qual.) PARTIAL	slide 5; no run converged
Vanilla fails to converge for $n \geq 8$		slide 6; $\mathcal{L} \approx 0.50$ at $n \geq 8$
Q-LINK gradient variance $\geq 10\times$ Vanilla for $n \geq 7$		slide 4: Adaptive $\geq 10\times$ at $n \geq 7$ Fixed $\sim 1-3\times$
Expressibility same order of magnitude as Table I	PASS	slide 3
Loss-landscape qualitative match to Fig. 6	PASS	slide 7

Score. 3 PASS, 1 PARTIAL, 1 MISS. The MISS is on absolute convergence speed, traced to a cost-function discrepancy in the reference repo.

Diagnosis — Why the Convergence Target Misses

Paper Eq. (2): $\hat{O}_{\text{paper}} = I - \frac{1}{n_{\text{data}}} \sum_i Z_i$ on the data register.

Reference repo (AARC-lab/DCAS_2026_QLINK / cost_function.py):

- Implements $1 - \frac{1}{n_{\text{data}}} \sum_i P(q_i = |0\rangle)$ — i.e. exactly *half* the paper's operator.
- Additional Q-LINK-specific bug: loop iterates $j \in [0, 2^{n_{\text{data}}})$ over a $2^{n_{\text{tot}}}$ -length probability vector. In TC big-endian, this projects the first data qubit onto $|0\rangle$ and replaces it with the messenger in the measured set:

$$\hat{O}_{\text{buggy}} = I - \frac{1}{n_{\text{data}}} \sum_{k=1}^{n_{\text{tot}}-1} P_0^{(q_0)} P_0^{(q_k)}.$$

- Unit test `test_buggy_observable_matches_literal_loop` pins this operator to the verbatim Python loop for $n_{\text{tot}} \in \{3, 4, 5\}$.

Effect on our numbers. Running the paper's *stated* cost is flat near the random-init state and SGD does not escape in 1500 iters. Running the *buggy* operator (replication of authors' actual code) reproduces the published trajectory variances at the order-of-magnitude level (slide 4) but inflates Q-LINK(F) by the $P_0^{(q_0)}$ projector floor.

Summary — Ours vs Theirs at a Glance

Metric	Paper headline	Our replication
Expressibility (Table I)	drops to 0 at $n \geq 9$	drops to 0 at $n \geq 9$ — match
Vanilla grad var. ($n = 10$)	1.18×10^{-7}	1.31×10^{-7} — match
Q-LINK(F) grad var. ($n = 10$)	9.62×10^{-7}	2.13×10^{-7} — $4.5\times$ low
Q-LINK(A) grad var. ($n = 10$)	7.27×10^{-6}	4.36×10^{-5} — $6\times$ high
Convergence ratio ($\eta_{\text{Q-LINK(F)}}$)	10.55	1.00 — never converged
Vanilla saturates at $n \geq 8$	yes	yes — match
Landscape ordering ($V > \text{Adap-}$ $\text{tive} > \text{Fixed}$)	yes	yes — match

Three quantitative quantities reproduce within an order of magnitude; three qualitative claims match cleanly. Convergence ratio is the lone miss, owned and diagnosed (slide 8). Repo: github.com/satyapratheektata/qmlink-replication.